

## Chapter 10

### Worksheet

#### Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

#### Worksheet for Chapter 10: Capstone Project and Future Vision

**Purpose:** Reinforce key concepts from Chapter 10 through reflective questions, practical activities, and goal-setting exercises to help students apply their learning from designing AI solutions, collaborating on projects, presenting ideas, reflecting on growth, and planning for future skills development.

#### Part 1: Designing an AI Solution for a Real-World Problem

**Objective:** Identify a problem and brainstorm an AI-based solution to address it.

- Think about your daily life or community. What's 1 problem you'd like to solve (e.g., "traffic delays to school")?
  - Problem: \_\_\_\_\_
- How could AI help solve this problem (e.g., "predict best routes using traffic data")? Write 1-2 sentences.
  - AI Solution Idea: \_\_\_\_\_
- Who would benefit from your solution (e.g., "students and parents")?
  - Target Group: \_\_\_\_\_

#### Part 2: Collaborating on a Group Tech Project

**Objective:** Reflect on teamwork dynamics and plan for effective collaboration.

- If you worked in a team, list 1 strength each member brought (e.g., "Ravi: great at research"). If solo, list 2 strengths you used.
  - Strength 1: \_\_\_\_\_
  - Strength 2: \_\_\_\_\_
- What was 1 challenge you faced while working together (e.g., "disagreement on app features")? How did you handle it or plan to next time?
  - Challenge & Solution: \_\_\_\_\_
- Suggest 1 way to improve team communication for future projects (e.g., "daily 5-min check-ins").
  - Idea: \_\_\_\_\_

#### Part 3: Presenting Ideas to Peers and Mentors

**Objective:** Prepare and refine skills for pitching your AI project idea.

- Write a 2-sentence "elevator pitch" for your AI solution (e.g., "Our app uses AI to spot bullying online. It keeps teens safe by alerting trusted adults.").
  - Pitch: \_\_\_\_\_

- What's 1 way you can make your presentation engaging (e.g., "start with a relatable story" or "use a funny image")?
  - Engagement Idea: \_\_\_\_\_
- Imagine you get feedback after presenting (e.g., "Add a feature for parents."). How would you respond to show you value it?
  - Response: \_\_\_\_\_

#### Part 4: Reflecting on Learning and Growth

**Objective:** Assess personal development and lessons learned from the capstone project.

- What's 1 new skill or idea about AI/tech you learned during this chapter (e.g., "AI needs data to work")?
  - New Learning: \_\_\_\_\_
- What was the toughest part of this project, and how did you overcome it (or plan to next time)?
  - Challenge & Approach: \_\_\_\_\_
- What are you most proud of from this project, and why (e.g., "Our team's creativity; we thought outside the box")?
  - Proud Moment: \_\_\_\_\_

#### Part 5: Creating a Personal Action Plan for Future Skills Development

**Objective:** Set actionable goals to continue growing tech and AI skills.

- List 1 tech skill you want to improve or learn (e.g., "basic coding in Python").
  - Skill Goal: \_\_\_\_\_
- Write a SMART goal for it (Specific, Measurable, Achievable, Relevant, Time-bound), e.g., "Complete 3 Python lessons on Codecademy in 1 month, 30 mins weekly."
  - SMART Goal: \_\_\_\_\_
- Name 1 resource to help (e.g., "free YouTube channel: Tech with Tim") and 1 time slot to work on it (e.g., "Saturday 4-4:30 PM").
  - Resource: \_\_\_\_\_
  - Time Slot: \_\_\_\_\_

#### Part 6: Quick Self-Reflection

**Objective:** Evaluate your mindset and readiness for future tech challenges after this chapter. Circle your answer for each question:

- How confident are you in using AI concepts to solve real problems after this project?
  - Not Confident / Somewhat Confident / Very Confident
- How prepared do you feel to collaborate on tech projects with others?
  - Not Prepared / Somewhat Prepared / Very Prepared
- How motivated are you to keep learning tech skills for your future?
  - Not Motivated / Somewhat Motivated / Very Motivated

#### Bonus Activity: My Future Vision Board Idea

**Objective:** Visualize your tech dreams and commit to skill-building goals. Sketch or describe 1 element of a "Future Vision Board" for your tech journey. Include a dream role or project (e.g., "AI app developer") and 1 skill you'll need (e.g., "machine learning").

- Vision Board Idea: \_\_\_\_\_ Set a reminder to start working

on your SMART goal by:

- Date/Time: \_\_\_\_\_